Technical Reproducibility Validation Report

ExpansionHunter

Version v5.0.0
Medical STR

Verification date	2026-11-06
Repository commit	9f5db1d4ab72016b2d93330b9792842d71582622
Result	**Runs as documented**
Reached	Runs + Plausible output
Permalink	https://strhub.app/verified/expansionhunter-med
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Runs as documented**
REACHED	Runs + Plausible output, 6/6 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Step 1: The Documented Run

As documented, it runs.

A third party installed the tool with the steps this commit states and ran the command it writes, and the output file that command names appeared. This says nothing about genotypes or about the forensic panel.

3. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Starts	The installed program answers --help in the built environment	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	expansionhunter-med.log-external.txt	

4. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · ExpansionHunter v5.0.0 · commit 9f5db1d
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( mkdir \
-p sample &&
ExpansionHunter \
--reads /data/in/input.bam \
--reference /data/ref/hg38.fa \
--variant-catalog /data/in/regions.json \
--output-prefix sample \
); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

5. Reproducibility Metadata

Tool	ExpansionHunter
Version	v5.0.0
Repository	https://github.com/Illumina/ExpansionHunter
Commit (pinned)	9f5db1d4ab72016b2d93330b9792842d71582622
Container OS	ubuntu-22.04
Marker panel	Medical STR
Verified on	2026-11-06 16:19:55 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42250919800

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

6. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

7. Output Content Evidence

Output file	sample.vcf
Format	VCF
Sequence records	51
Distinct STR loci	51
Total reads	27,496
Max depth (single locus)	1103

8. Verification Data

The repository ships example data (4 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 51 disease-associated STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878, medical STR slice (female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	AFF2 · AFF3 · AR · ARX · ATN1 · ATXN1 · ATXN10 · ATXN2 · ATXN3 · ATXN7 · ATXN8OS · BEAN1 · C9orf72 · CACNA1A · CBL · CNBP · COMP · CSTB · DAB1 · DIP2B · DMPK · FMR1 · FOXL2 · FXN · GIPC1 · GLS · HOXA13 · HOXD13 · HTT · JPH3 · LRP12 · MARCHF6 · NIPA1 · NOP56 · NOTCH2NLC · NUTM2B-AS1 · PABPN1 · PHOX2B · PPP2R2B · RAPGEF2 · RFC1 · RUNX2 · SAMD12 · SOX3 · STARD7 · TBP · TCF4 · TNRC6A · XYLT1 · YEATS2 · ZIC2
Regions BED	Provided by STRhub

9. Verification Matrix

N/A	Run on repository test data	—
	Run on reference dataset	Illumina BAM (hg38), NA12878, medical STR slice (female)
	README check (4/5)	4/5 items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

11. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's install instructions (release_binary), at the pinned commit.
– Input: the README documents BAM (line 15); this run used STRhub's illumina-bam-hg38-med reference data.
– Panel: the loci file the documented command reads, shipped in the repository, lists coordinates inside STRhub's medical STR slice and none inside the forensic one; this run used the medical slice of the same sample (illumina-bam-hg38-med).
– The output directory the command writes into (sample) was created first.
– 1 input path(s) in the README command replaced with /data/in/input.bam.
– 1 reference FASTA path(s) replaced with /data/ref/hg38.fa.
– Run command: the one documented in docs/03_Usage.md (line 15), rewritten to STRhub's mounts; everything it created was captured as output.
– Regions: STRhub's ready-made eh_catalog file for the dataset's panel loci (STRhub records the eh_catalog layout for a program of this name).

12. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

13. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

◆ **Runs end-to-end**

ExpansionHunter v5.0.0 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

◆ **Produces plausible output**

The tool generated a structurally valid VCF output file with genotype calls across 51 target forensic STR loci, with a maximum read depth of 1103 at SAMD12.

◆ **Verdict: Runs**

The tool installed and its run produced its documented output.

◆ **Repository test data not used**

The repository ships example data (4 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878, medical STR slice (female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth (1/2)

ExpansionHunter v5.0.0 · verified 2026-11-06

LOCUS	READ DEPTH		LOCUS	READ DEPTH	
SAMD12	1103		**PHOX2B**	584	
RAPGEF2	1015		**ZIC2**	560	
AFF2	876		**ARX**	560	
YEATS2	864		**CBL**	548	
MARCHF6	855		**BEAN1**	547	
DAB1	834		**HOXD13**	510	
STARD7	776		**HTT**	498	
ATXN2	775		**TCF4**	484	
ATXN10	730		**RUNX2**	470	
AR	724		**C9orf72**	445	
TNRC6A	691		**JPH3**	416	
TBP	689		**ATN1**	414	
ATXN3	642		**NIPA1**	414	
FMR1	624		**NOTCH2NLC**	400	
SOX3	610		**PPP2R2B**	398	
RFC1	608		**GIPC1**	396	
ATXN1	605		**ATXN8OS**	386	
FOXL2	600		**XYLT1**	374	
HOXA13	592		**CNBP**	364	
PABPN1	588		**NUTM2B-AS1**	363	

Appendix: Detected Markers with Read Depth (2/2)

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LOCUS	READ DEPTH		LOCUS	READ DEPTH	
CSTB	363	■■■■■	**GLS**	313	■■■■
NOP56	358	■■■■■	**AFF3**	310	■■■■
DMPK	347	■■■■■	**COMP**	303	■■■■
CACNA1A	345	■■■■■	**DIP2B**	301	■■■■
FXN	330	■■■■	**ATXN7**	266	■■■
LRP12	328	■■■■			